

The Methodological Pitfalls of Data Analysis

Lecture Study Notes on Statistical Sins and Machine Learning Vulnerabilities

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Abstract

These lecture study notes synthesize core principles from Prof. John Guttag's MIT OpenCourseWare lecture on machine learning classification and statistical pitfalls. The main thesis establishes that the mathematical validity of any analytical or predictive model depends fundamentally on underlying data integrity and strict adherence to structural assumptions. Performing flawless mathematical calculations on corrupted or non-representative data merely accelerates the generation of false conclusions, rendering bad data analysis significantly more dangerous than no analysis at all. These notes systematically examine historical cases of data corruption, visual graphics manipulation, sampling biases, and high-dimensional machine learning vulnerabilities including multicollinearity, regularization penalties (L_1 vs. L_2), probability thresholding, and Receiver Operating Characteristic (ROC) metrics.

1. Introduction: The Principle of Data Integrity (GIGO)

The foundational axiom of quantitative computing is Garbage In, Garbage Out (GIGO). Tracing back to Charles Babbage's 1864 reflections on his mechanical Analytical Engine, calculating machinery possesses no internal ability to rectify erroneous inputs into valid outputs. Babbage recalled being asked by members of Parliament whether putting wrong figures into the machine would yield right answers, noting that he was unable to apprehend the confusion of ideas that could provoke such a question. In modern computational settings, applying sophisticated statistical algorithms or machine learning models to corrupt data produces mathematically precise falsehoods.

A prime historical illustration of GIGO occurred during the 1840 U.S. Census, when Secretary of State John Calhoun utilized census summaries to argue that freed Black individuals in Northern states suffered from dramatically higher rates of mental illness than enslaved individuals in the South. Former President John Quincy Adams exposed glaring errors in the census, demonstrating that several Northern towns had mistakenly recorded mentally ill white residents as Black or listed non-existent populations. When confronted, Calhoun asserted that the dataset contained so many errors that they would simply balance each other out across the aggregate, illustrating a persistent misconception regarding statistical aggregation.

The mathematical invalidity of Calhoun's defense rests on the structural distinction between random and systematic errors. Let y_i represent the true underlying value for individual i , and let x_i be the recorded value in the presence of measurement error e_i , such that $x_i = y_i + e_i$. If errors are independent and identically distributed according to an unbiased Gaussian distribution

$e_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$ with probability density function:

$$f(e_i) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{e_i^2}{2\sigma^2}\right) \quad (1)$$

the expected error value is zero, $\mathbb{E}[e_i] = 0$. By the Strong Law of Large Numbers, the sample mean error $\bar{e} = \frac{1}{N} \sum_{i=1}^N e_i$ converges almost surely to zero as sample size $N \rightarrow \infty$, guaranteeing that $\mathbb{E}[\bar{x}] = \mathbb{E}[\bar{y}]$. However, when errors are systematic or politically corrupted, the error distribution has a non-zero mean $\mathbb{E}[e_i] = \mu_e \neq 0$. Consequently, the expected value of the sample mean becomes $\mathbb{E}[\bar{x}] = \mu_y + \mu_e \neq \mu_y$. Increasing the sample size N fails to diminish systematic bias μ_e , proving that bad data cannot self-correct through aggregation.

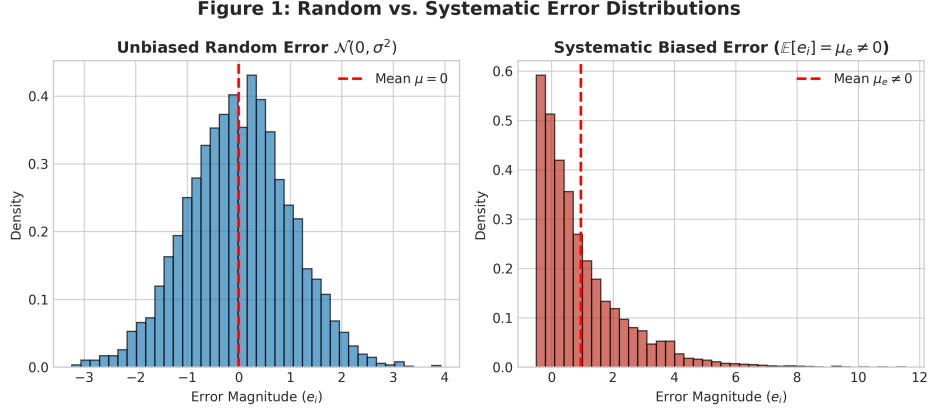


Figure 1: Random vs. Systematic Error Distributions. The left panel shows an unbiased Gaussian distribution where positive and negative errors cancel out in aggregate ($\mathbb{E}[e] = 0$). The right panel illustrates a skewed systematic error distribution shifted by $\mu_e \neq 0$, proving that aggregation cannot eliminate systematic data corruption.

2. The Illusion of Summary Statistics and Visual Manipulations

Analysts frequently commit statistical errors by evaluating datasets exclusively through numerical summary metrics such as the mean, variance, and correlation coefficient without visualizing raw data distributions. In 1973, statistician Francis Anscombe demonstrated the danger of this oversight by constructing four distinct datasets ($N = 11$ observations each), known as Anscombe's Quartet. Each dataset possesses identical first- and second-order summary statistics: mean of x ($\bar{x} = 9.0$), sample variance of x ($s_x^2 = 11.0$), mean of y ($\bar{y} = 7.50$), sample variance of y ($s_y^2 \approx 4.125$), correlation coefficient ($r_{xy} \approx 0.816$), and identical fitted linear regression equations $\hat{y} = 3.0 + 0.5x$ with $R^2 \approx 0.67$ and residual sum of squares $SS_{\text{res}} \approx 13.75$. Despite possessing identical numerical summaries, visual inspection reveals completely disparate geometric structures, including standard linear noise, non-linear quadratic curves, tight linear trends with vertical outliers, and high-leverage extreme points.

Beyond summary statistics, graphical representations are routinely manipulated to deceive viewers through truncated vertical baselines, selective scaling, and incommensurate variable comparisons. Consider two group metrics $y_1 = 4.0$ and $y_2 = 4.1$, yielding a true proportional difference of:

$$\Delta_{\text{true}} = \frac{y_2 - y_1}{y_1} = 2.5\% \quad (2)$$

If an analyst truncates the vertical baseline by displaying the y -axis starting at $y_0 = 3.9$, the visually perceived height ratio becomes:

$$R_{\text{visual}} = \frac{y_2 - y_0}{y_1 - y_0} = \frac{4.1 - 3.9}{4.0 - 3.9} = \frac{0.2}{0.1} = 2.0 \quad (3)$$

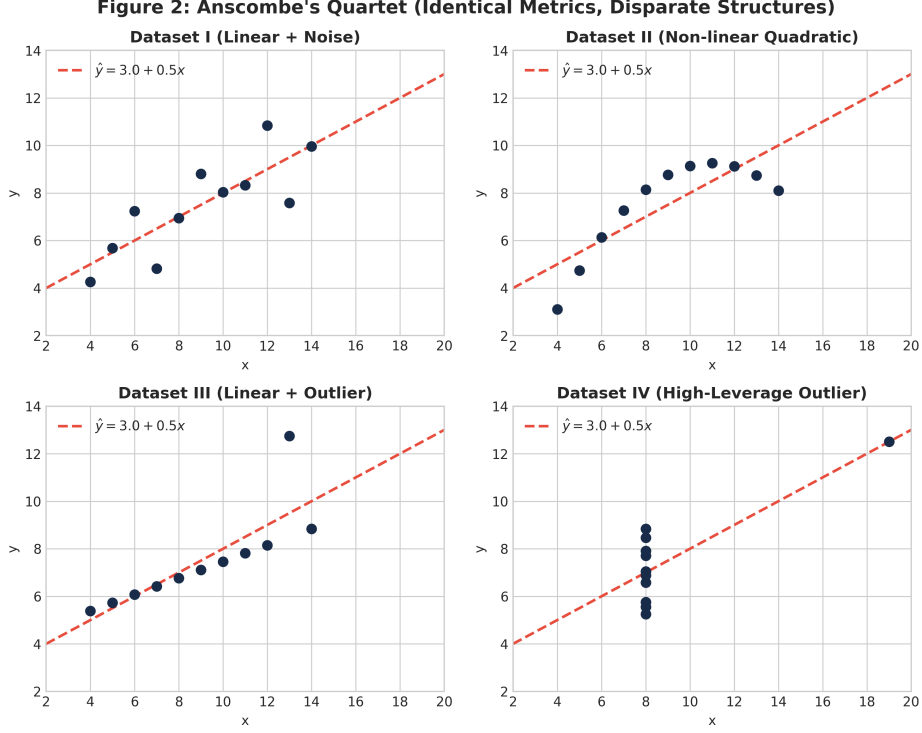


Figure 2: Anscombe's Quartet. A grid of four scatter plots sharing identical numerical summary metrics and fitted linear regression lines ($\hat{y} = 3.0 + 0.5x$), demonstrating that statistics about data do not specify the underlying data distribution.

representing an artificial visual exaggeration factor of $E = \frac{2.0}{1.025} \approx 1.95$, or a 95% visual distortion. As highlighted in Prof. Guttag's lecture, broadcast media graphics often employ truncated baselines alongside non-comparable metrics, such as comparing the total number of individuals in a household receiving any welfare benefit against individual full-time jobholders, creating misinformed public perceptions.

3. Sampling Biases and the Violation of Core Assumptions

Classical inferential statistics depend strictly on the assumption that sample observations are drawn randomly and independently from the target population distribution $F(x)$, expressed as $X_i \stackrel{\text{i.i.d.}}{\sim} F(x)$. Under pure random sampling, the Central Limit Theorem dictates that as sample size $N \rightarrow \infty$, the sample mean $\bar{X}$ converges to a normal distribution:

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{N}} \xrightarrow{d} \mathcal{N}(0, 1) \quad (4)$$

with standard error $\text{SE}(\bar{X}) = \frac{\sigma}{\sqrt{N}}$.

When sampling is dictated by convenience, non-response, or survival filters, observations are instead drawn from a conditional distribution $F(x | S = 1)$, where S represents a selection indicator variable governed by feature vector $\mathbf{Z}$. When selection probability depends on features $P(S = 1 | \mathbf{Z}) \neq P(S = 1)$, the expected sample value deviates from the true population mean:

$$\mathbb{E}[X | S = 1] = \mu + \text{Bias} \neq \mu \quad (5)$$

Consequently, classical standard error formulas and hypothesis test p -values become mathematically invalid.

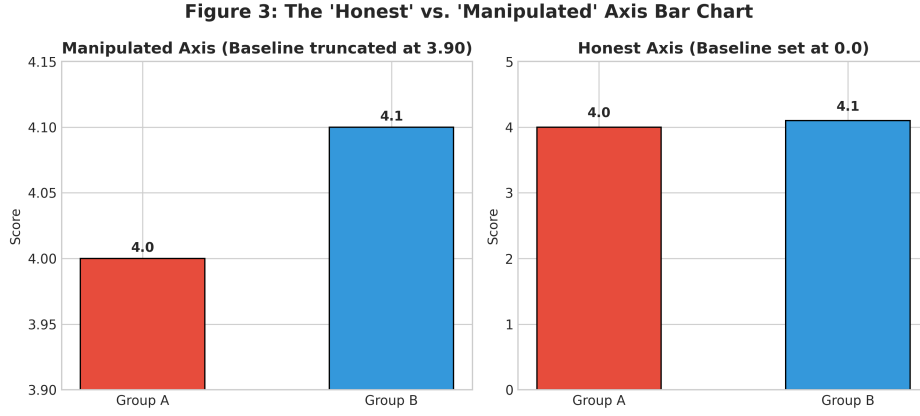


Figure 3: The "Honest" vs. "Manipulated" Axis Bar Chart. The left chart truncates the y -axis baseline to $[3.90, 4.05]$, artificially magnifying a 2.5% score variance into a 2 : 1 visual ratio. The right chart displays an honest zero baseline (0.0 to 5.0), accurately showing minimal variation.

A prominent operational example is World War II survivor bias analyzed by statistician Abraham Wald. Military analysts examined returning aircraft to identify flak hole patterns, intending to reinforce damaged areas with extra armor. Wald recognized that the sample suffered from conditional filtering:

$$P(\text{Observed Damage at Location } L) = P(\text{Hit at } L \mid \text{Plane Survived}) \quad (6)$$

Aircraft hit in critical engine modules failed to survive and were excluded from the airfield sample ($S = 0$). Thus, high damage density on returning aircraft indicated non-lethal impact zones, proving that armor should be added to areas showing no bullet holes on returning planes. Similar sampling filters occur in modern surveys, such as political landline polling excluding mobile users or course evaluations suffering from non-response bias due to early student dropouts.

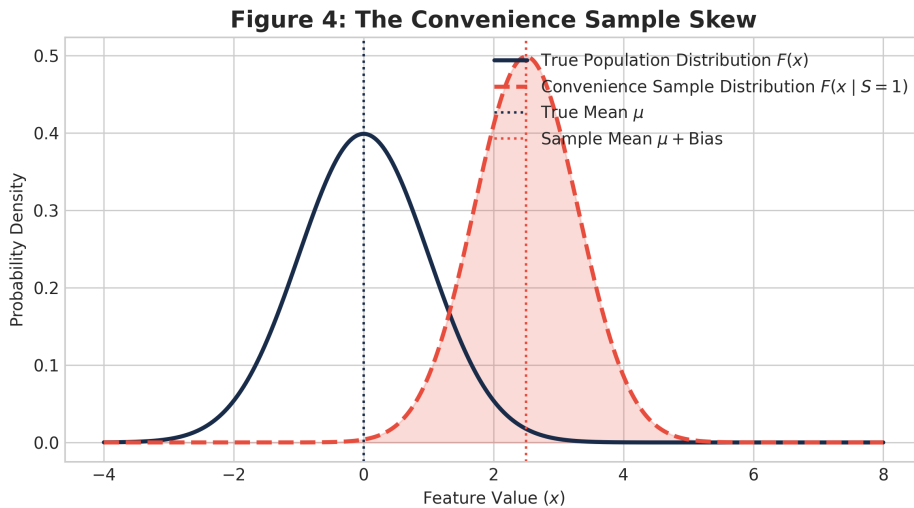


Figure 4: The Convenience Sample Skew. A probability density plot contrasting the true population distribution $F(x)$ against a biased convenience sample distribution $F(x \mid S = 1)$, showing that expanding sample size N in a biased sample merely increases precision around an incorrect mean.

4. Machine Learning Vulnerabilities: Over-interpretation and Thresholds

Machine learning classifiers provide predictive utility, but interpreting their inner parameter weights requires extreme caution due to data partitioning variance and feature correlations. Evaluating models across different random 80/20 train/test data splits demonstrates that accuracy varies continuously. Declaring algorithm superiority based on a single train/test evaluation split is methodologically flawed.

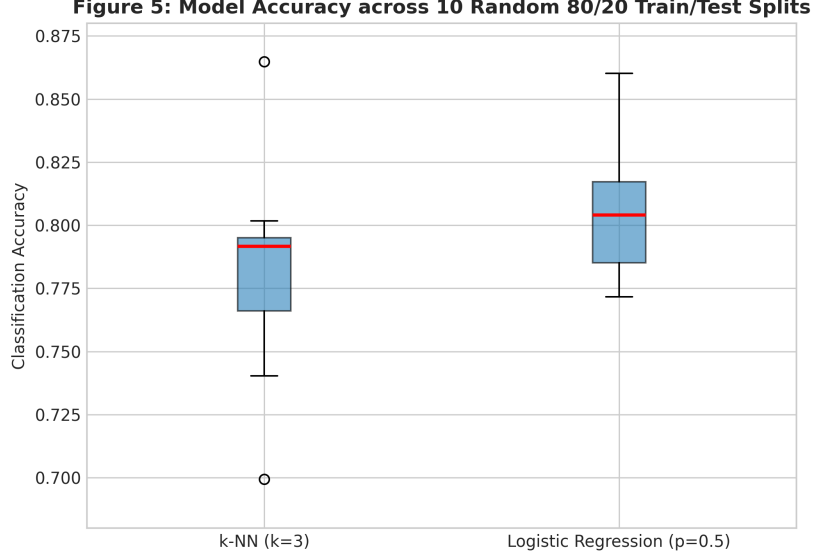


Figure 5: Model Comparison Boxplots (k -NN vs. Logistic Regression). Performance distributions across 10 random 80/20 train/test splits for k -Nearest Neighbors ($k = 3$) and Logistic Regression ($p = 0.5$), highlighting model variance across random data partitions.

High-dimensional modeling introduces severe overfitting when the number of input features d approaches the number of sample observations N . As $d \rightarrow N$, models fit random sample noise rather than underlying signal, driving training accuracy toward 100% while test set generalization accuracy drops sharply.

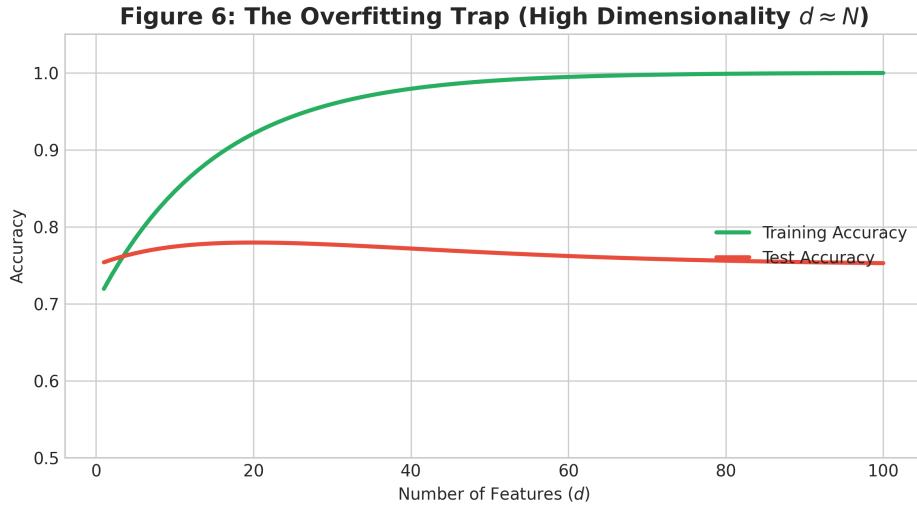


Figure 6: The Overfitting Trap (High Dimensionality $d \approx N$). Training accuracy approaches 100% while test accuracy plummets as the number of features approaches sample size.

For a binary classification task $Y \in \{0, 1\}$ given feature vector $\mathbf{x} = [x_1, \dots, x_d]^T$, logistic regression estimates conditional probability $p = P(Y = 1 \mid \mathbf{x})$ using the sigmoid function $\sigma(z) = \frac{1}{1+e^{-z}}$.

Applying logit transformation models log-odds as a linear parameter combination:

$$\text{logit}(p) = \ln\left(\frac{p}{1-p}\right) = \mathbf{w}^T \mathbf{x} + b = w_0 + w_1 x_1 + \cdots + w_d x_d \quad (7)$$

Interpreting individual feature weights w_j in isolation is invalid when inputs exhibit multicollinearity. In Titanic passenger survival modeling, binary cabin class dummy variables satisfy the exact linear invariant $c_1 + c_2 + c_3 = 1 \implies c_1 = 1 - c_2 - c_3$. Including all dummy features alongside an intercept causes design matrix X to lose full rank ($\text{rank}(X) < d+1$), making covariance matrix $X^T X$ singular. Parameter variance scales with inverse covariance:

$$\text{Var}(\hat{\mathbf{w}}) = \sigma^2 (X^T X)^{-1} \rightarrow \infty \quad \text{as} \quad \det(X^T X) \rightarrow 0 \quad (8)$$

Eliminating one collinear feature (e.g., removing c_1) resolves matrix singularity without altering overall predictive accuracy, but wildly shifts remaining feature parameter weights, as summarized in Table 1.

Table 1: Impact of Feature Elimination on Logistic Regression Parameter Weights (Titanic Dataset)

Model Specification	Class 1 (c_1)	Class 2 (c_2)	Class 3 (c_3)	Age	Male Gender
Full Feature Set (c_1, c_2, c_3)	+1.60	+0.46	-0.80	-0.02	-2.40
Reduced Set (Eliminated c_1)	—	-1.25	-2.80	-0.03	-2.35

To control high-dimensional overfitting, regularized objective functions penalize weight magnitudes through regularizer term $\Omega(\mathbf{w})$ added to loss function $\mathcal{L}$:

$$\min_{\mathbf{w}} \sum_{i=1}^N \mathcal{L}(y_i, \sigma(\mathbf{w}^T \mathbf{x}_i)) + \Omega(\mathbf{w}) \quad (9)$$

L_2 Ridge regularization uses penalty $\Omega_{L_2}(\mathbf{w}) = \lambda \|\mathbf{w}\|_2^2 = \lambda \sum_{j=1}^d w_j^2$, shrinking weights continuously toward zero and spreading coefficients across correlated features. L_1 Lasso regularization uses penalty $\Omega_{L_1}(\mathbf{w}) = \lambda \|\mathbf{w}\|_1 = \lambda \sum_{j=1}^d |w_j|$, which drives redundant feature weights to exactly zero due to non-differentiable penalty corners at $w_j = 0$.

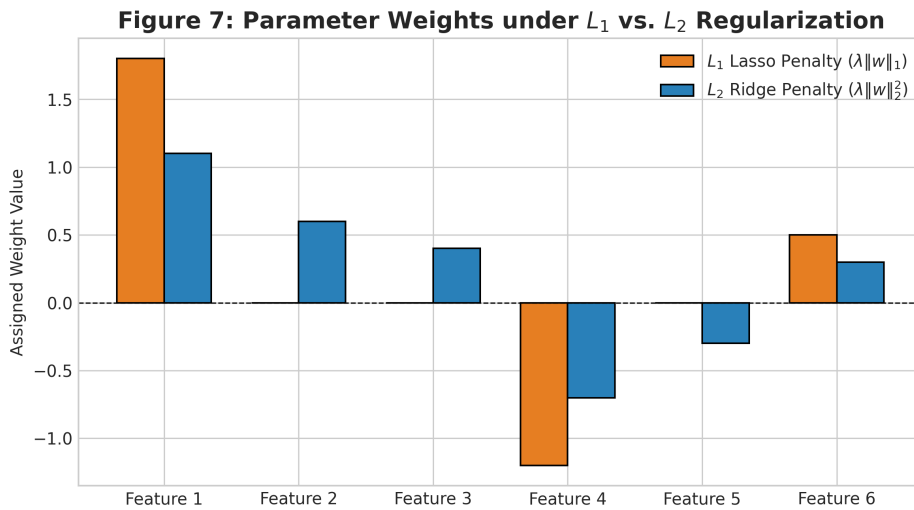


Figure 7: Parameter Weights under L_1 vs. L_2 Regularization. L_1 regularization imposes feature sparsity by setting less impactful collinear weights to exactly zero, whereas L_2 spreads weights across correlated inputs.

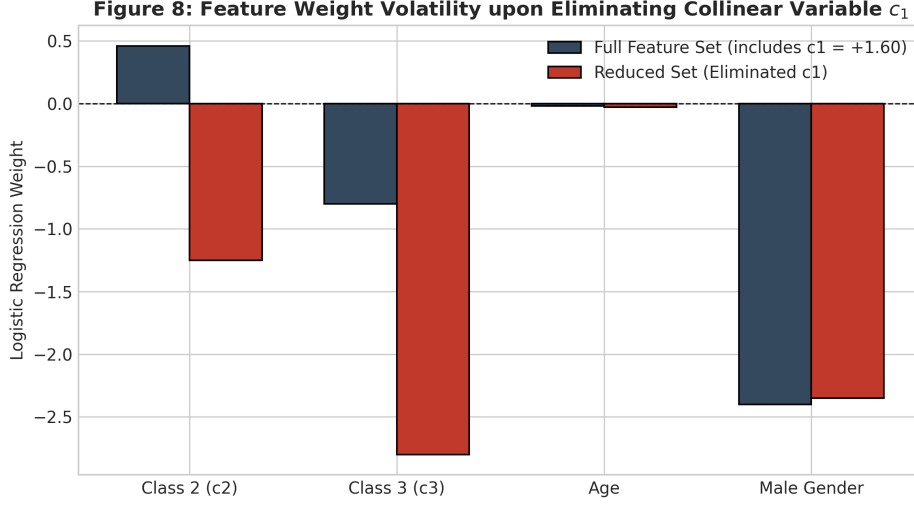


Figure 8: Feature Weight Volatility Bar Chart. Visualizing parameter shifts from Table 1, demonstrating that eliminating collinear feature c_1 causes c_2 and c_3 weights to shift from positive values to large negative values.

Binary classification assigns positive label $\hat{Y} = 1$ when predicted probability meets or exceeds decision cutoff threshold $P(Y = 1 \mid \mathbf{x}) \geq p$. Classification performance metrics are defined via the Confusion Matrix:

$$\text{Sensitivity (TPR)} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad \text{Specificity (TNR)} = \frac{\text{TN}}{\text{TN} + \text{FP}}, \quad \text{FPR} = \frac{\text{FP}}{\text{TN} + \text{FP}} \quad (10)$$

Adjusting cutoff p shifts operational performance. Setting a conservative threshold $p = 0.9$ requires high model certainty to predict survival, maximizing specificity while depressing sensitivity. In business settings, such as insurance underwriting, setting p too conservatively to eliminate risk prevents writing viable policies, harming revenue.

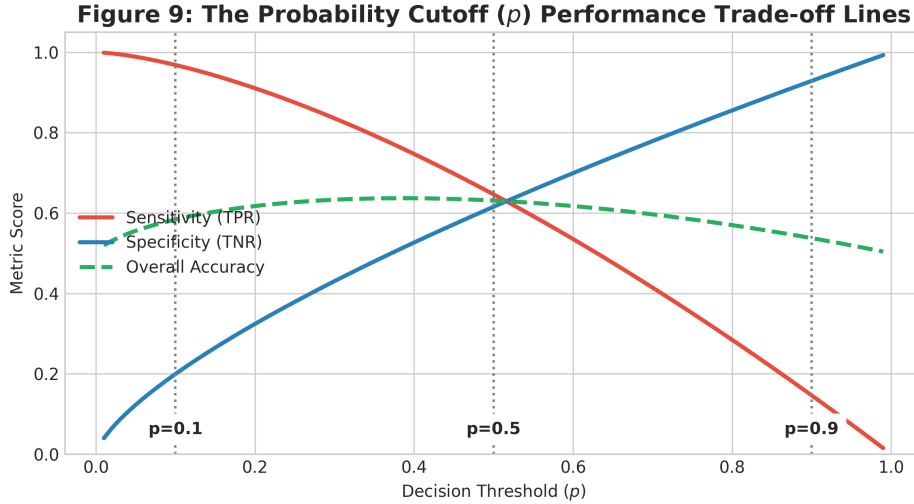


Figure 9: The Probability Cutoff (p) Trade-off Lines. Accuracy, Sensitivity, and Specificity plotted across decision cutoff thresholds $p \in [0.0, 1.0]$.

Evaluating classifiers across all operational cutoffs is achieved using the Receiver Operating Characteristic (ROC) curve, which plots Sensitivity (TPR) against $1 - \text{Specificity}$ (FPR). The Area Under the ROC Curve (AUROC) provides an overall performance index:

$$\text{AUROC} = \int_0^1 \text{TPR}(p) d(\text{FPR}(p)) = P(f(\mathbf{x}_+) > f(\mathbf{x}_-)) \quad (11)$$

AUROC equals the probability that a randomly selected positive instance $\mathbf{x}_+$ scores higher than a randomly selected negative instance $\mathbf{x}_-$. A random classifier achieves AUROC = 0.5, whereas a perfect classifier achieves AUROC = 1.0.

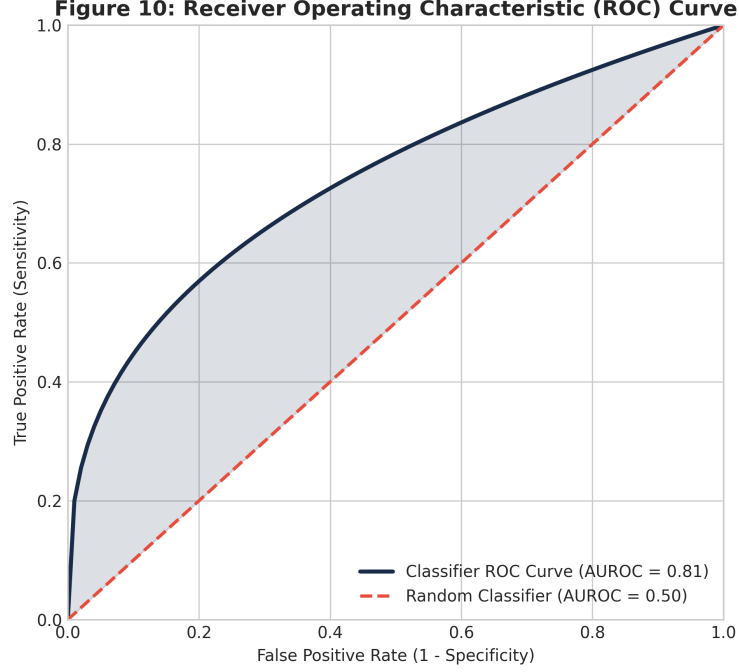


Figure 10: Receiver Operating Characteristic (ROC) Curve. Plotting True Positive Rate against False Positive Rate (1 – Specificity), comparing classifier performance against a random $y = x$ baseline (AUROC = 0.5).

Evaluating whether an estimated AUROC significantly exceeds random chance (0.5) relies on the test statistic:

$$z = \frac{\text{AUROC} - 0.5}{\text{SE}(\text{AUROC})} \quad (12)$$

where standard error scales inversely with sample size, $\text{SE}(\text{AUROC}) \propto \frac{1}{\sqrt{N}}$. As sample size expands $N \rightarrow \infty$, standard error approaches zero $\text{SE} \rightarrow 0$. Consequently, on massive datasets ($N = 1,000,000$), a weak classifier with AUROC = 0.52 achieves high statistical significance ($p < 0.001$) despite offering zero practical real-world utility.

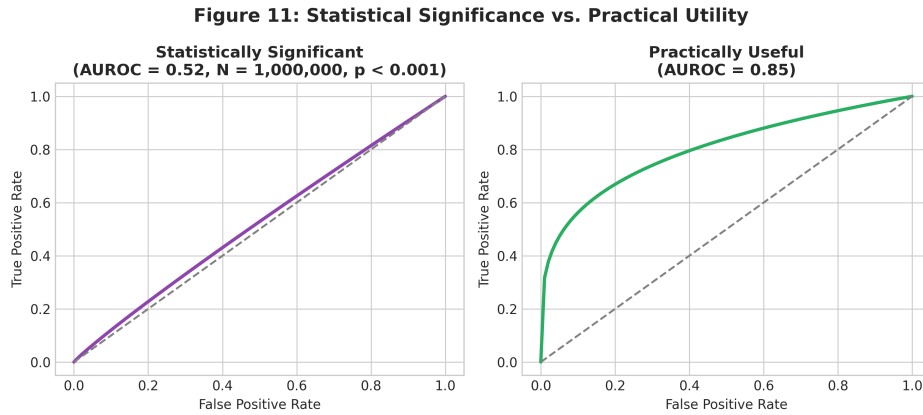


Figure 11: Statistical Significance vs. Practical Utility. The left panel shows a statistically significant classifier on a massive dataset ($N = 1,000,000$, $p < 0.001$) with negligible utility (AUROC = 0.52). The right panel shows a practically useful classifier (AUROC = 0.85).

5. Conclusion

As statistician Darrell Huff famously noted, if one cannot prove what they want to prove, demonstrating something else and pretending they are identical often deceives casual observers during the collision of statistics with the human mind. Maintaining quantitative integrity requires evaluating data collection mechanisms before running models, visually plotting raw distributions prior to computing summary metrics, verifying axis scales, validating random sampling assumptions, accounting for feature collinearity when analyzing model weights, and distinguishing statistical significance from practical utility.

References

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- [2] Anscombe, F. J. (1973). Graphs in Statistical Analysis. *The American Statistician*, 27(1), 17–21.
- [3] Huff, D. (1954). *How to Lie with Statistics*. W. W. Norton & Company.